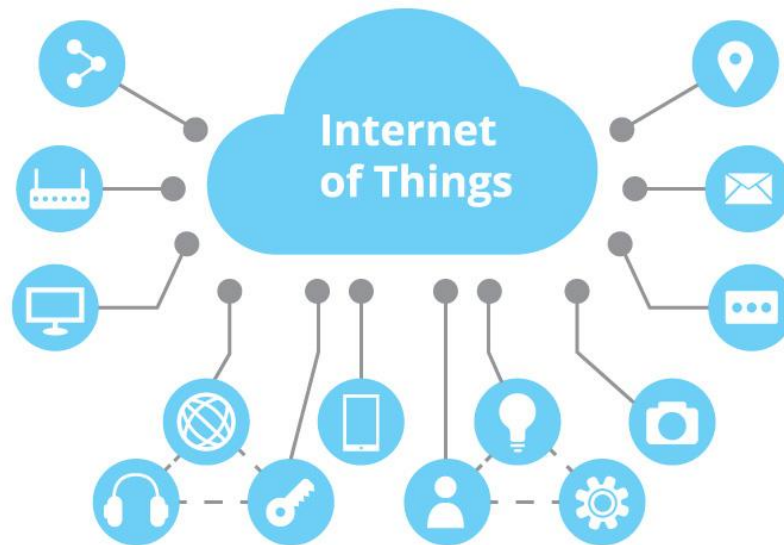


Internet of Things (IoT)



Chapter 1: Introduction to IoT System

Nguyen Tan No

03/2025

Contact

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Target

- Describe the concepts and components in the IoT System.
- Design, developed hardware, and build applications in the IoT System.
- Program, monitor, and control applications and components of IoT System.
- Manage, analyze, and process data within IoT systems.
- Develop several commonly used IoT systems in practical applications.

Ebook

- 1: Monika Bhatnagar, N. Rajendran, R. Priyadarshini, Ashok (2023), Internet of things (IoT) systems and Applications. NTL Publication.
- 2: Peter Waher (2015). Learning Internet of Things. Packt Publishing.
- 3: Jamil Y. Khan, Mehmet R. Yuce (2019). Internet of Thing (IoT) Systems and Applications. Jennu Stanford Publishing.

Evaluate

- **Attendance:** **10%**
- **Discussion:** **10%**
- **Midterm Presentation:** **30%_Simulation**
- **Final Project:** **50%_Experiment**

Course description

This course provides students with the necessary knowledge and skills to design, implement, and manage IoT systems. The curriculum covers topics such as sensors, connected devices, communication protocols, IoT application programming and development, as well as security and data analysis.

Students will learn how to use embedded control platforms like Arduino, STM32, ESP32, etc., to build practical IoT applications, ranging from smart homes to applications in agriculture, healthcare management, and more.

The course aims to equip students with the ability to develop effective and secure IoT solutions, while also gaining an understanding of emerging technological trends in the field.

Course description

1 Chapter 1: Introduction to IoT System

1.1. Overview of the IoT System

1.2. Concepts in LAN Network System, Ethernet

1.3. Communication Protocols: Zigbee, Lora

2 Chapter 2: Description of Hardware Devices in the IoT System

2.1. Embedded Systems

2.2. Local Communication Protocols: ADC, I2C, SPI, UART, LoRa, 3G

3 Chapter 3: Sensors and Actuators in the IoT System

3.1. Sensors

3.2. Actuators

Course description

4 Chapter 4: Building an IoT System

4.1. Connecting Sensor Communication Hardware

4.2. Programming Actuator Control

4.3. Integrating Control Programming for Systems

5 Chapter 5: Design of Monitoring and Control System

5.1. Design of Local Network Monitoring Interface

5.2. Design of Internet-Based Monitoring and Control Interface

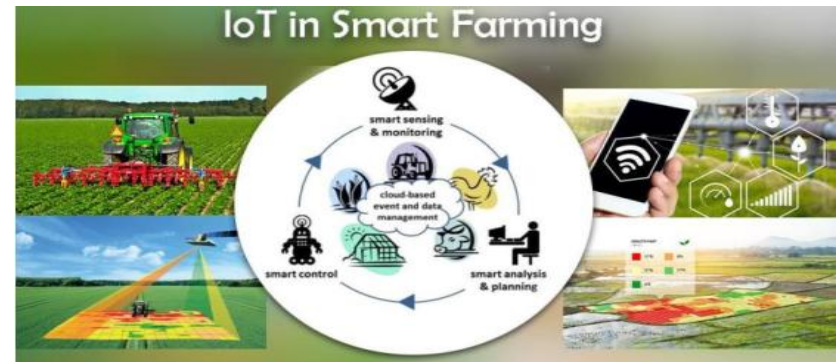
5.3. Network Security

Course description

Chapter 6: Building IoT Applications

6.1. Smart Agriculture

1. Controller (Wifi)(*)
2. pH Sensor
3. Relay Module/Fan(*)
4. Temp/Humidity Sensor (*)
5. Light/PWM(*)
6. Lora
7. Solar cell/Battery



Course description

Chapter 6: Building IoT Applications (continued) 6.2. Smart Homes - Smart Cities

1. Controller (Wifi) (*)
2. Zigbee Module(*)
3. RFID Module(*)
4. Light/PWM(*)
5. Voice Sensor
6. Motion detect

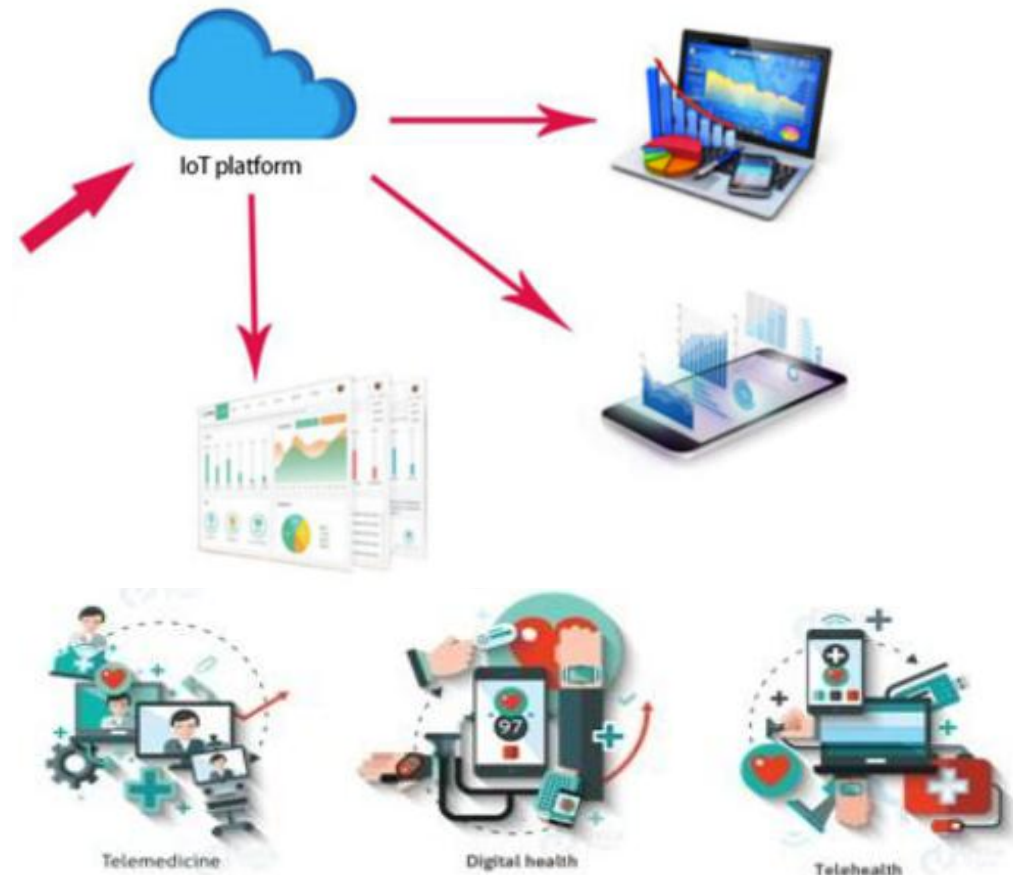


Course description

Chapter 6: Building IoT Applications (continued)

6.3. Healthcare

1. Controller (Wifi)(*)
2. Heart Rate/SpO2(*)
3. Temperature Sensor
4. Motion Detect Sensor
5. Accelerator Sensor(*)
- 6.....



Course description

Building IoT Applications (continued) 6.4. Transportation

1. Controller (Wifi)(*)
2. GPS/3G Module(*)
3. Ultrasonic Sensor(*)
4. Light Sensor
5. UHF RFID(*)
6. Camera



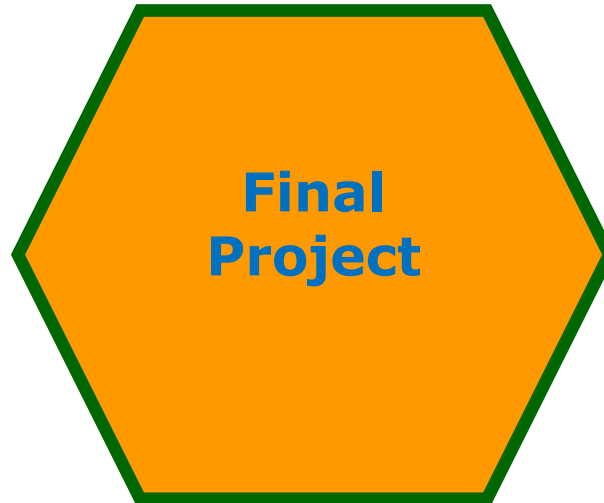
Final Project

Chapter 6: Building IoT Applications 6.1. Smart Agriculture

1. Controller (Wifi)^(*)
2. pH Sensor
3. Relay Module/Fan^(*)
4. Temp/Humidity Sensor^(*)
5. Light/PWM^(*)
6. Lora
7. Solar cell/Battery

Chapter 6: Building IoT Applications 6.2. Smart Homes - Smart Cities

1. Controller (Wifi)^(*)
2. Zigbee Module^(*)
3. RFID Module^(*)
4. Light/PWM^(*)
5. Voice Sensor
6. Motion detect



Chapter 6: Building IoT Applications 6.3. Healthcare

1. Controller (Wifi)^(*)
2. Heart Rate/SpO2^(*)
3. Temperature Sensor
4. Motion Detect Sensor
5. Accelerator Sensor^(*)
- 6....

Building IoT Applications 6.4. Transportation

1. Controller (Wifi)^(*)
2. GPS/3G Module^(*)
3. Ultrasonic Sensor^(*)
4. Light Sensor
5. UHF RFID^(*)
6. Camera

Time Schedule

1	2	3	4	5	6	7	8	9
Overview of IoT System + Software + Basic Program ming	Overview Embbded System + Communic ation protocol	Sensors + Actutuators	Building an IoT System	Monitorin g and Control System	Midterm Presentation			

Chapter 1

Introduction to IoT Systems

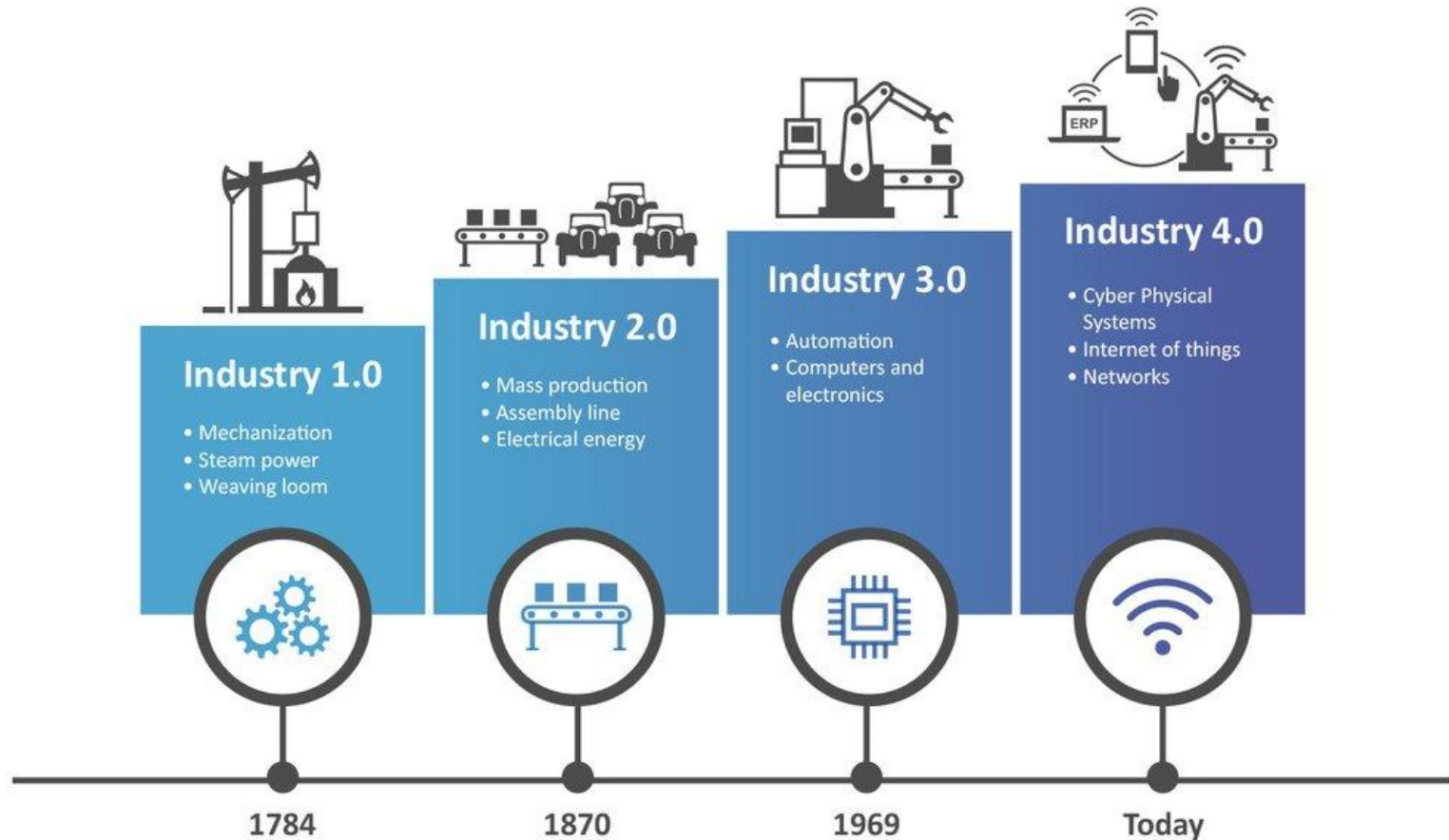
Chapter 1. Introduction to IoT Systems

1.1. Overview of the IoT System.

**1.2. Concepts in LAN Network System,
Ethernet System.**

1.3. Communication Protocol: Zigbee, Lora.

1.1. Overview of the IoT System.



1.1. Overview of the IoT System.

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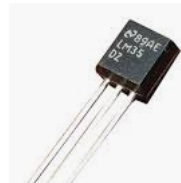


INDUSTRY 3.0 Automatization

Introduced electronics and computers to replace manual work by stand-alone robotic systems

1969 First programmable logic controller (PLC), Modicon 084

Sensors



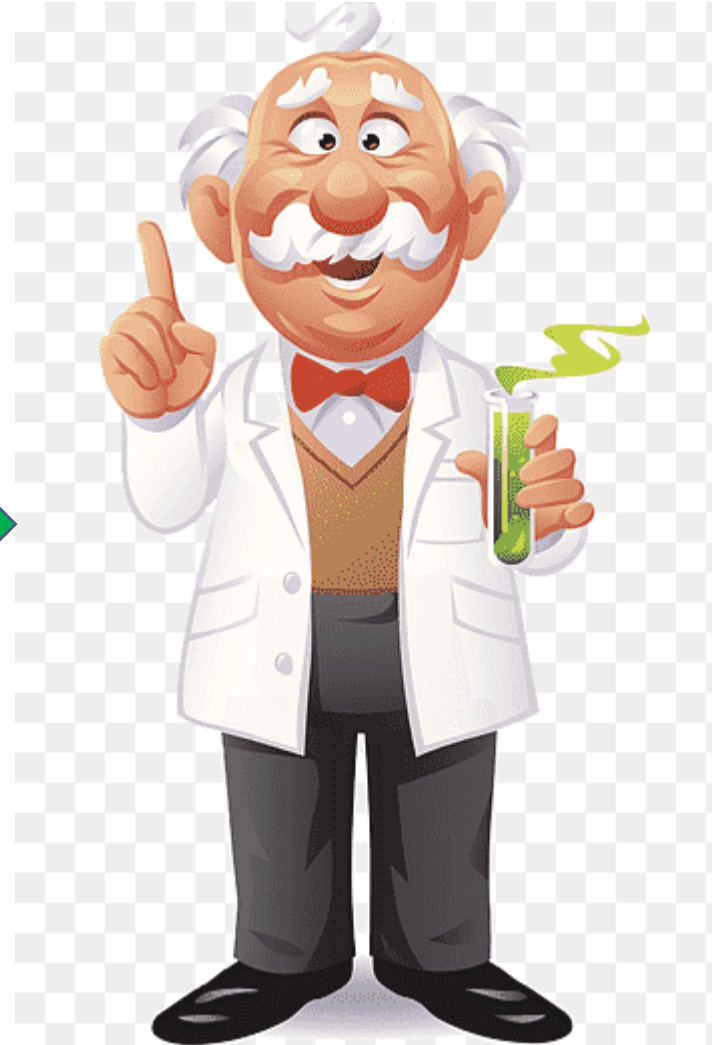
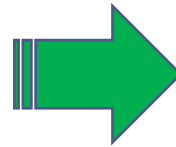
Controllers



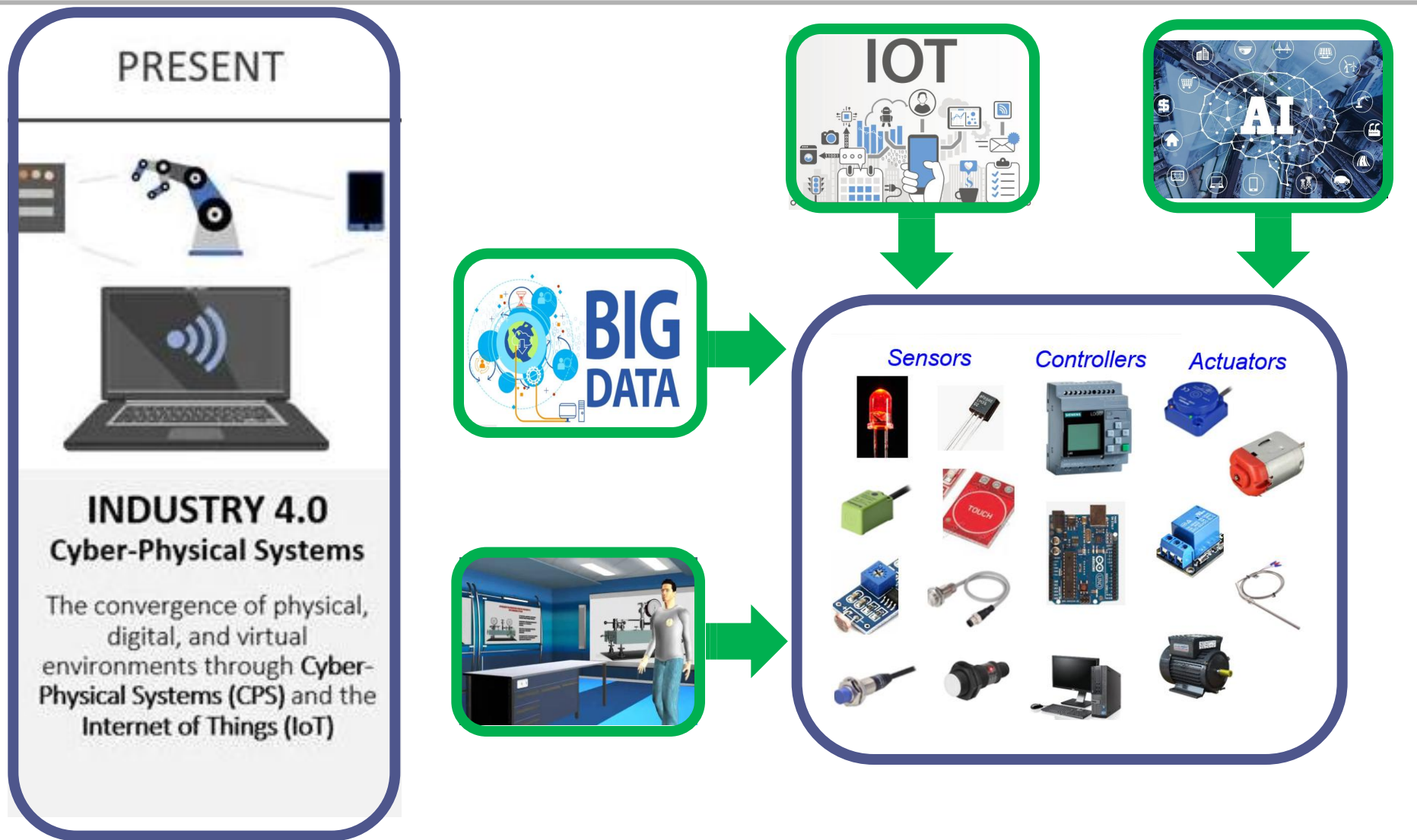
Actuators



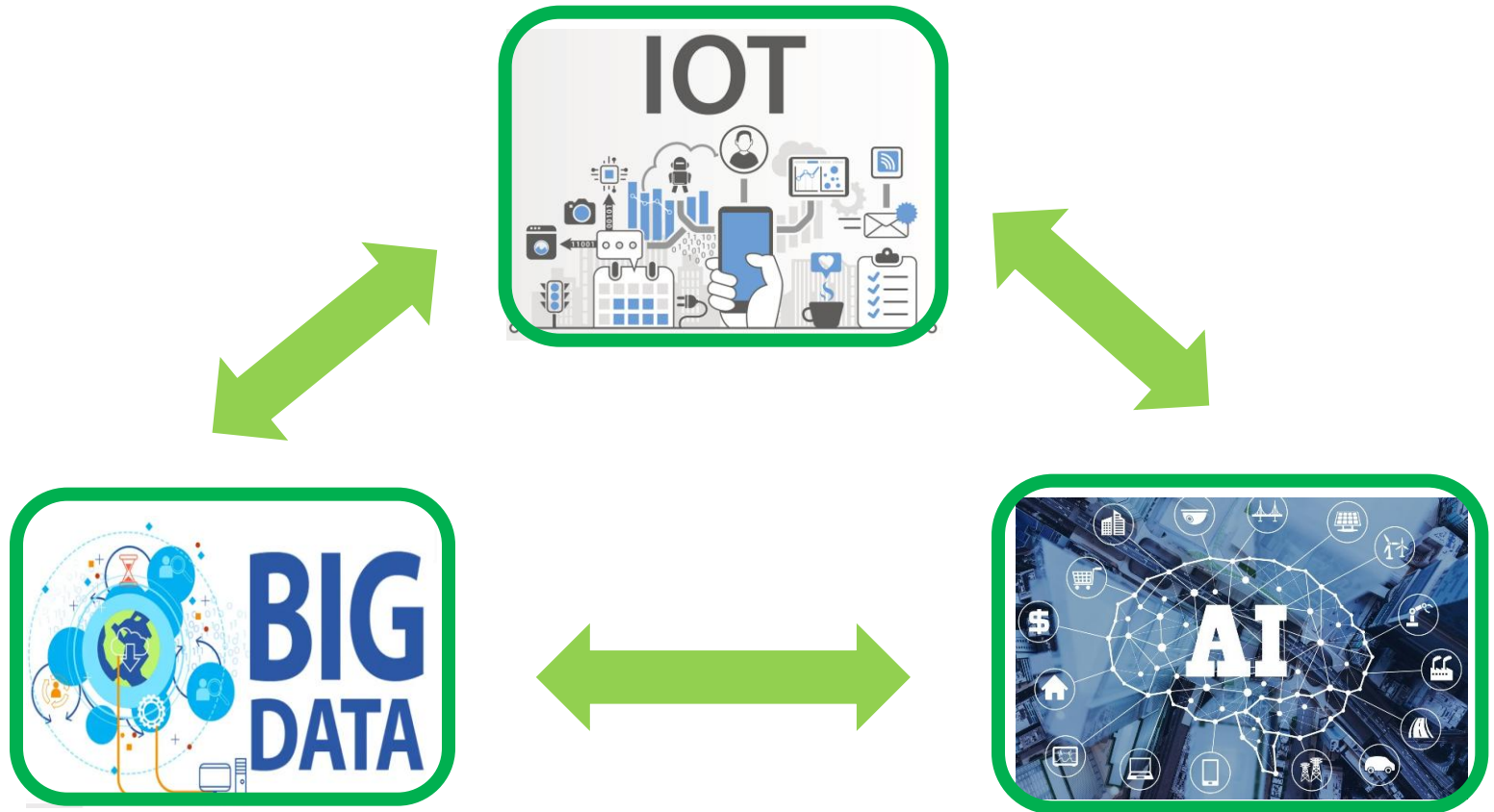
1.1. Overview of the IoT System.



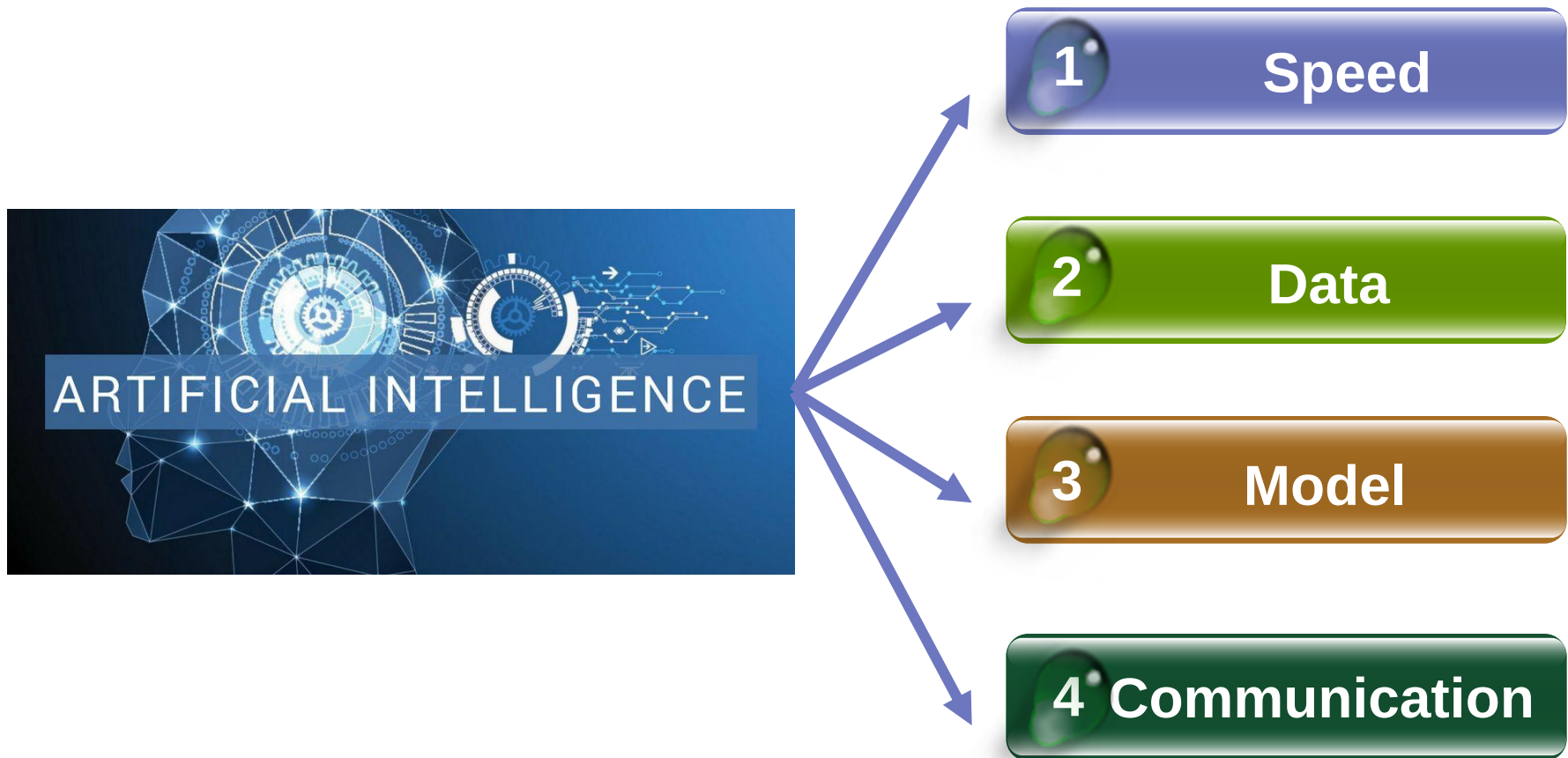
1.1. Overview of the IoT System.



1.1. Overview of the IoT System.



1.1. Overview of the IoT System.



1.1. Overview of the IoT System.

1.“The Internet of Things (IoT) is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.”

2.The Internet of Things is actually a pretty simple concept . It means taking all the physical places and things in the world and connecting them to the internet

3.IoT (Internet of Things) systems are networks of physical devices, vehicles, appliances, and other objects embedded with sensors, software, and network connectivity, enabling them to collect, exchange, and act on data, creating a vast interconnected ecosystem

4.When something is connected to the internet, that means that it can send information or receive information, or both . This ability to send and/or receive information makes things “smart .

1.1. Overview of the IoT System.

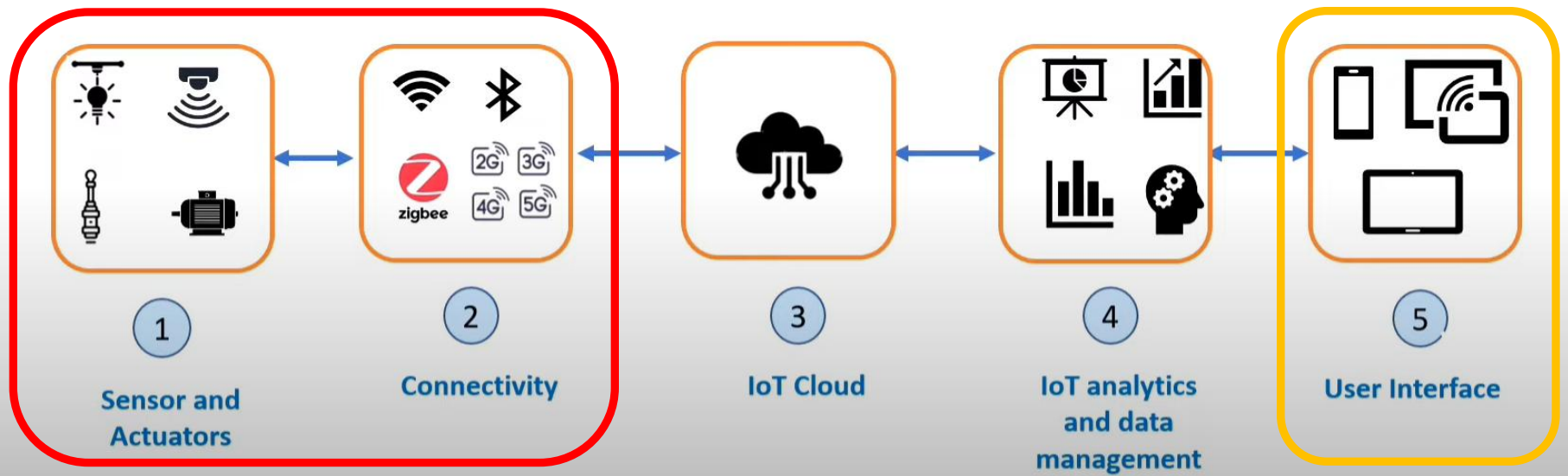


1.1. Overview of the IoT System.

Thing



1. Things that collect information and then send it
2. Things that receive information and then act on it
3. Things that do both



1.1. Overview of the IoT System.



1.1. Overview of the IoT System.

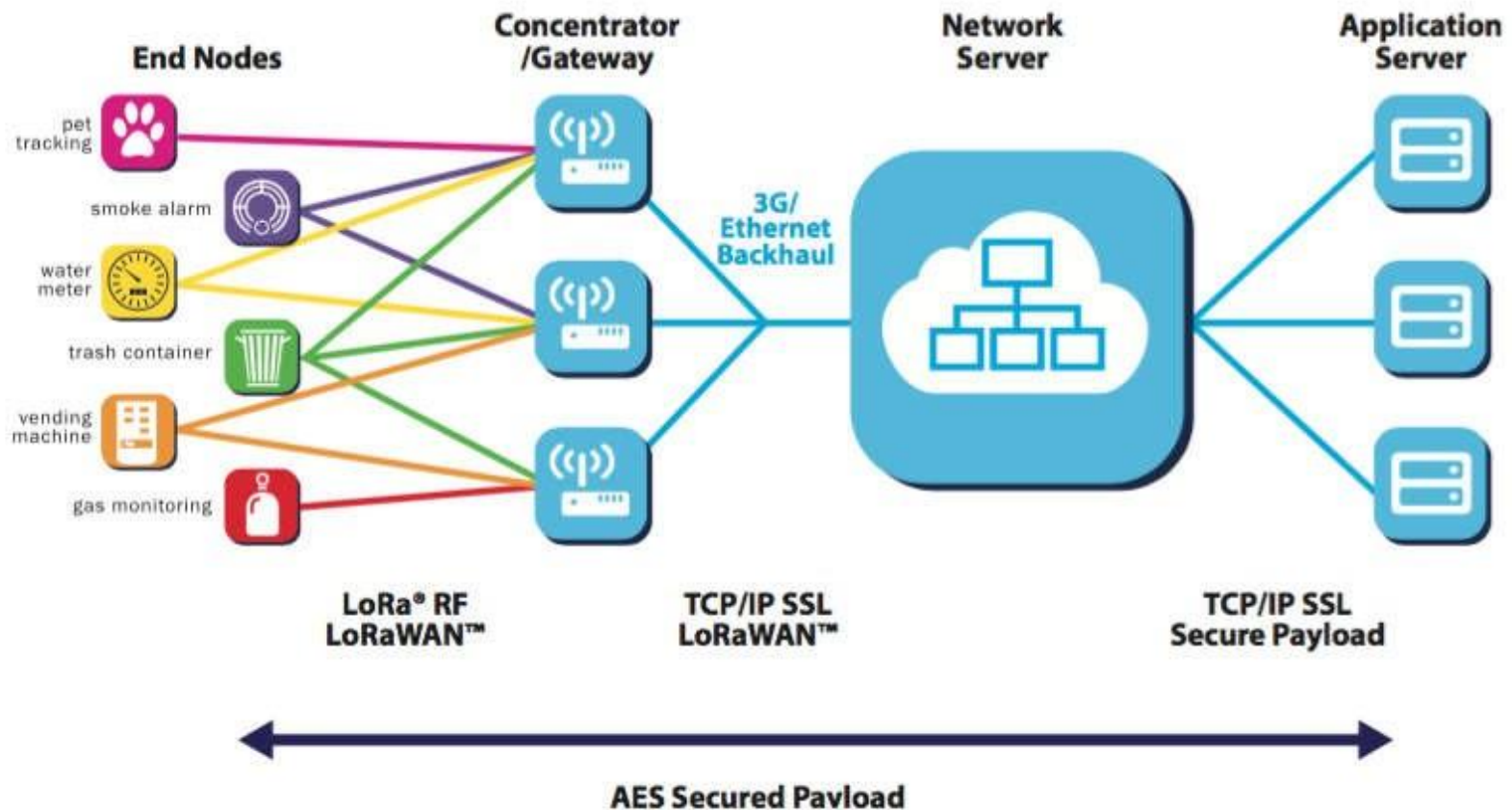


1.1. Overview of the IoT System.



1. Microcontroller
2. Sensors/Actuators
3. Connectivity Module

1.2. Concepts in LAN Network System, Ethernet System.



1.2. Concepts in LAN Network System, Ethernet System.

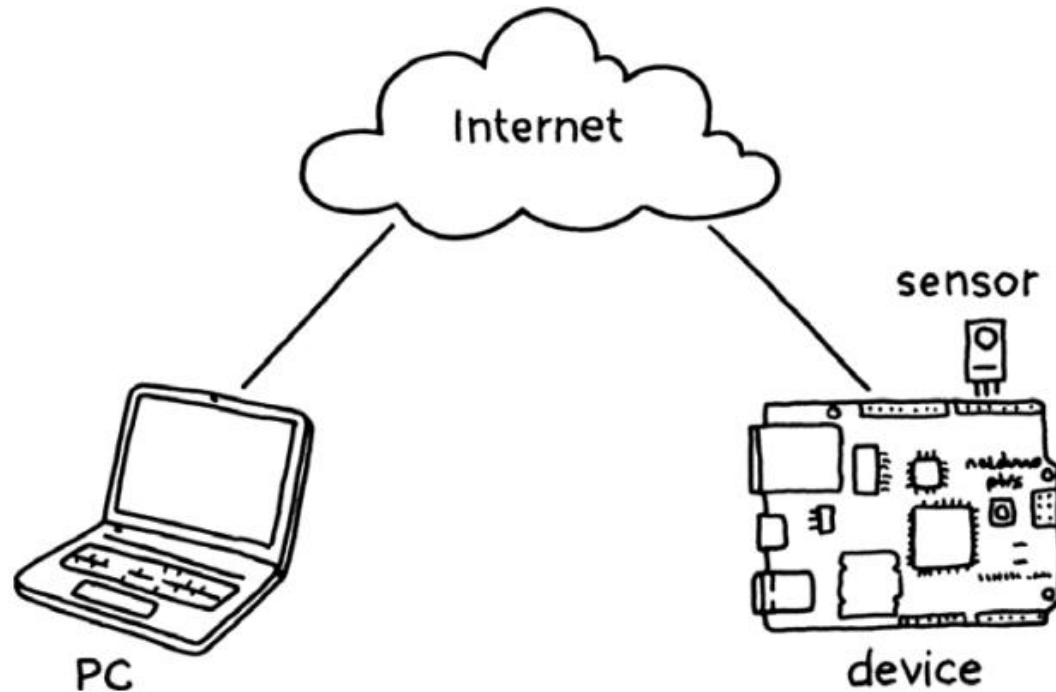
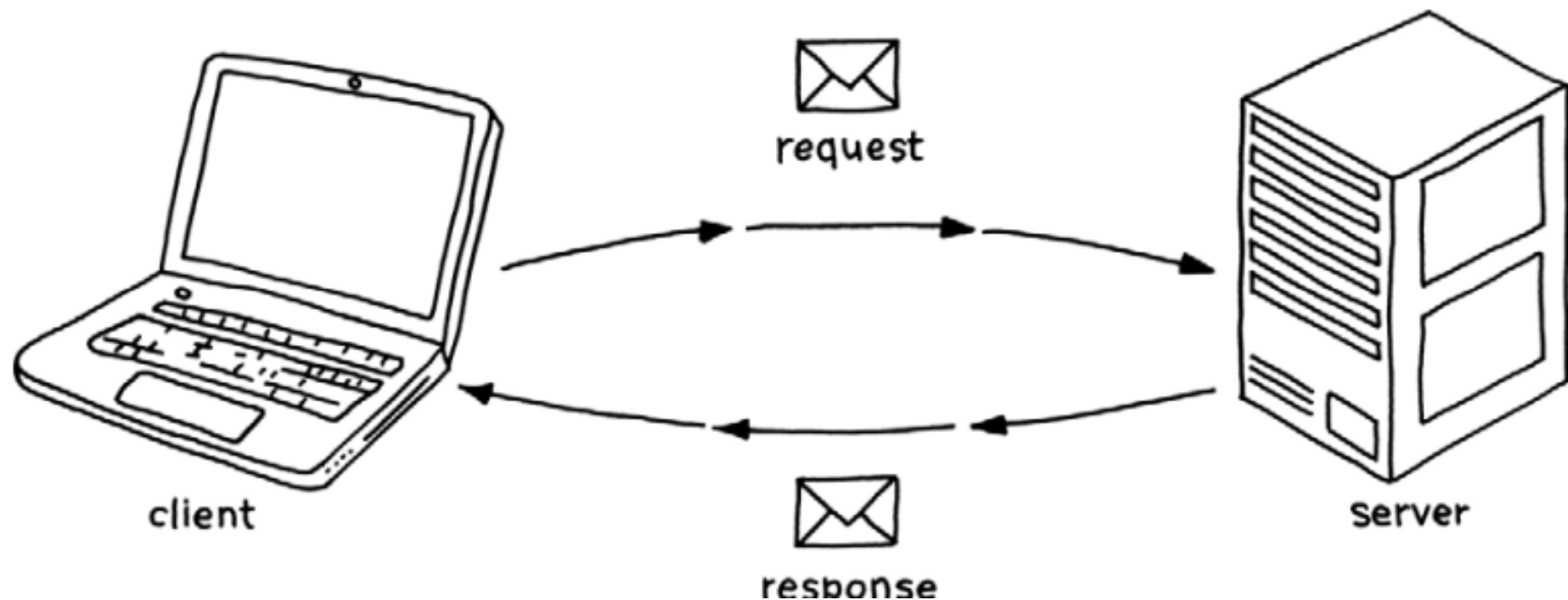


Figure 4-1. A PC and a device connected through the Internet

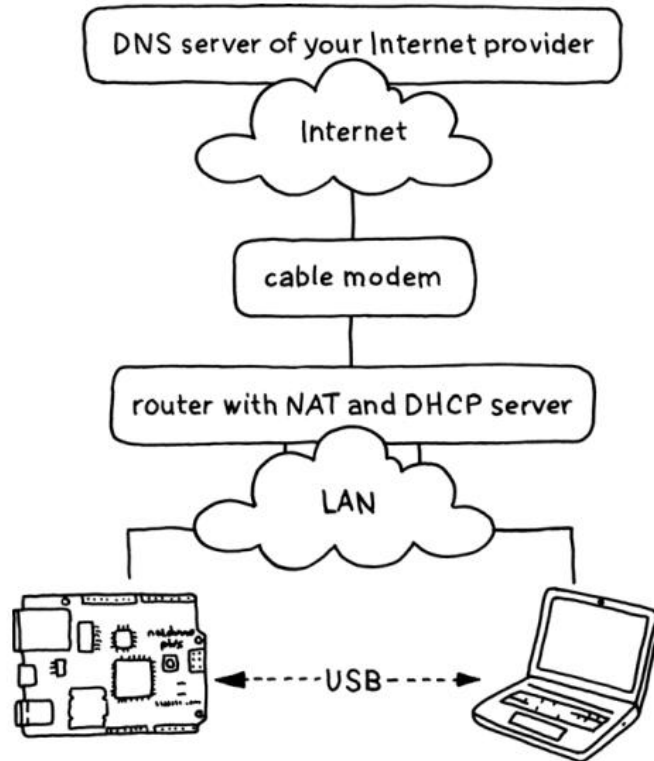
The *TCP/IP* protocol is the key Internet protocol for such communication scenarios. It enables the transfer of byte streams between two computers in either direction. For example, using the *TCP/IP* protocol, the device in Figure 4-1 may periodically deliver temperature measurements to a program running on the PC.

1.2. Concepts in LAN Network System, Ethernet System.

The most important standard protocol by far is the *Hypertext Transfer Protocol* (HTTP), the protocol of the World Wide Web. HTTP describes how a client interacts with a server, by sending *request messages* and receiving *response messages* over TCP/IP, as diagrammed in Figure 4-2.



1.2. Concepts in LAN Network System, Ethernet System.



A router typically implements the *Dynamic Host Configuration Protocol* (DHCP). This protocol allows your development PC, your Netduino Plus, and other devices to automatically obtain *Internet addresses* (e.g., 192.168.0.3 for the PC, and 192.168.0.4 for the

1.3. Communication Protocol: Lora

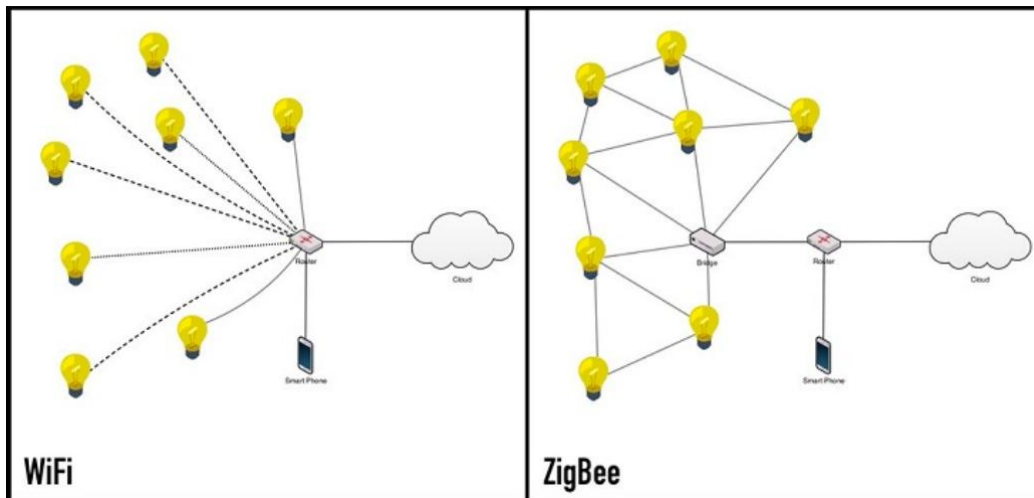
LoRa (Long Range) - A wireless communication technology designed for low-power, long-range applications, often used in the Internet of Things (IoT). It's based on chirp spread spectrum modulation and is widely used for connecting sensors and devices over long distances with minimal energy consumption.



LoRa module

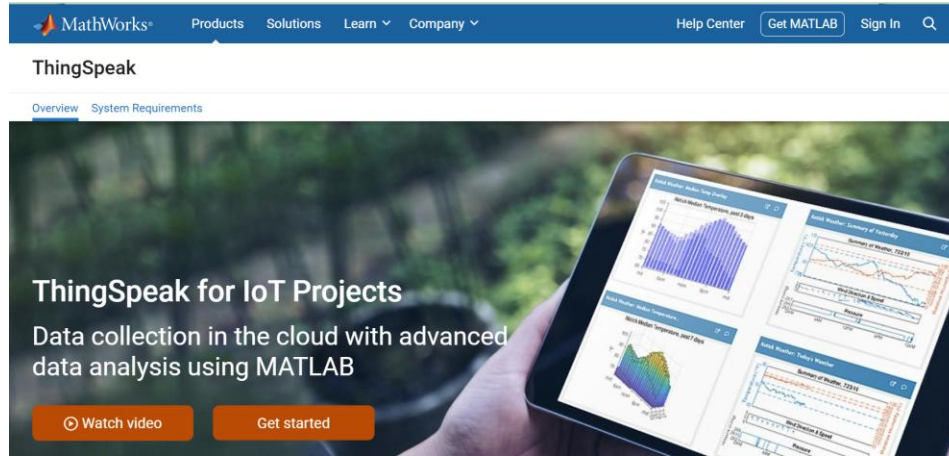
1.3. Communication Protocol: Zigbee,

Zigbee is a wireless protocol used by everything from IoT devices such as light bulbs, sockets, plugs, motion sensors and other home automation devices to smart energy, electric vehicle charging and industrial building controls that communicate with each other over a network



ZigBee Module

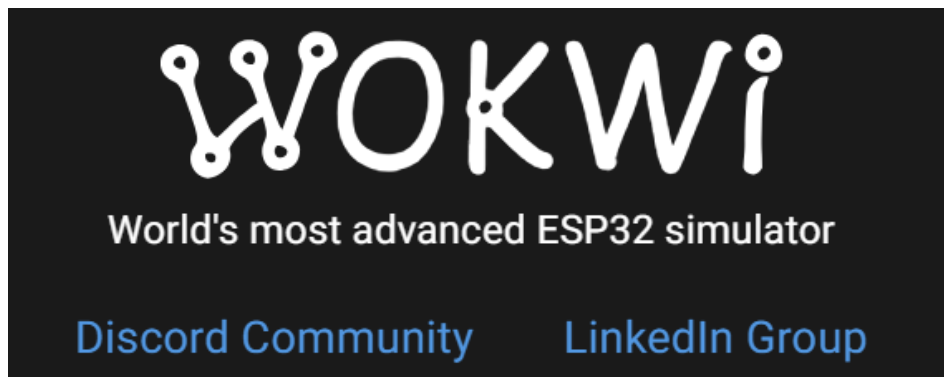
IoT Simulation Softwares



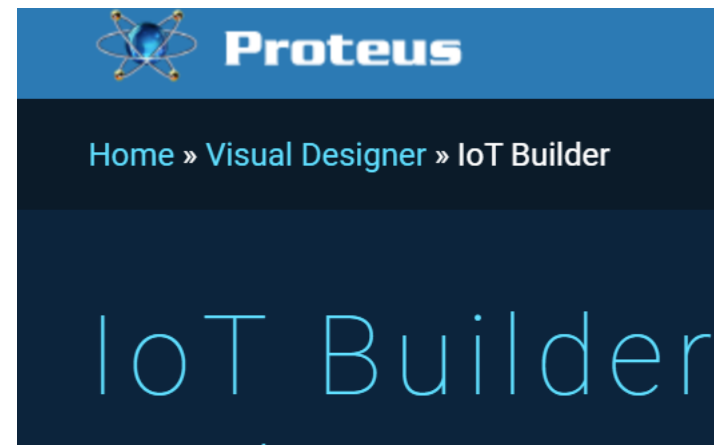
Thinkspeak/Matlab



Tinkercad



WokWi



IoT Builder

IoT System Developing Software



Visual Studio Code



Python



Visual Studio



Main Controller



Arduino



Esp32



STM32F



Raspberry

Thank for attention